

Graph Representation Learning

Abstract

Oversmoothing and over-squashing are two types of failures of information propagation in graph neural networks (GNNs): the first destroys signal essentially by too much mixing at depth, the latter by too little signal surviving a topological bottleneck. Two methods target them separately - residual (skip) connections for oversmoothing (as tested in one of our practicals) and graph rewiring for over-squashing. This miniproject studies whether these remedies complement each other or whether fixing one failure mode worsens the other. Using two controlled synthetic datasets (a stochastic block model for oversmoothing, a barbell signal-propagation task for over-squashing) and a 2×2 design over $\{\text{plain}, \text{skip}\} \times \{\text{original}, \text{rewired}\}$, evaluated across five seeds, we find a (fairly interesting but in the end not surprising) trade-off: curvature-based rewiring - as anticipated - fixes over-squashing but worsens oversmoothing, while skip connections mitigate oversmoothing + moderate over-squashing but fail on longer bottlenecks. Only the combination is robust on both axes. The mechanism is checked with Jacobian-sensitivity and Ollivier-Ricci curvature metrics.

Code: https://anonymous.4open.science/r/Graph_Representation_Learning_Miniproject-832F

1 Introduction

Message-passing GNNs propagate information by essentially repeatedly aggregating neighbour features. This single mechanism underlies two well-documented and opposite pathologies of information propagation:

Oversmoothing: as the depth grows, repeated averaging drives all node representations toward a common vector, so deeper networks become less discriminative [1, 2, 3].

Over-squashing: when a task requires information to travel through a structural bottleneck, an "exponentially growing receptive field" is compressed into a fixed-width vector and long-range signal is lost [4, 5, 6].

The two remedies presented in the lecture are usually studied in isolation. *Skip connections* [7] add a node's previous-layer representation back at each layer, counteracting oversmoothing. *Graph rewiring* [5] alters the input topology to relieve bottlenecks; Stochastic Discrete Ricci Flow (SDRF) adds edges at the most negatively curved (bottleneck) locations. Because the two pathologies could be called opposites, a natural and - to my knowledge - underexamined question is whether the two fixes interfere with each other.

Research question. *Do skip connections and graph rewiring complement each other in addressing oversmoothing and over-squashing, or does fixing the one worsen the other?* This matters because by mitigating one of the issues, researchers/practitioners could in theory worsen the other one without noticing right away.

2 Methods

Approach and goals. Each pathology is isolated in a synthetic graph where its severity is controllable via its parameters, and looked into via a 2×2 grid: $\text{model} \in \{\text{GCN}, \text{SkipGCN}\}$

and topology $\in \{\text{original}, \text{rewired}\}$, resulting in four cases: GCN, SKIPGCN, GCN+REWIRE, SKIP+REWIRE. The goal is to read off which mitigation technique helps or hurts which pathology.

Models. Both models stack K basic GCNConv layers (hidden width 64, ReLU, dropout 0.3). SKIPGCN uses the in Practical-3 introduced residual update $\mathbf{H}^{(t)} = \text{ReLU}(\text{GCNConv}(\mathbf{H}^{(t-1)})) + \text{proj}(\mathbf{H}^{(t-1)})$, where proj is a linear map when widths differ and identity otherwise.

Graph A: oversmoothing (SBM). A stochastic block model with 4 communities of 50 nodes ($p_{\text{intra}}=0.7$, $p_{\text{inter}}=0.02$, 16 random Gaussian features); node task = classify community. Dense intra-community connections make the repeated averaging collapse the node embeddings.

Graph B: over-squashing (barbell). Two 5-cliques joined by a bridge of length b (Fig. 1). A binary signal is out at a sourcenode in the left clique; all other nodes carry only random noise. The model must predict the signal only at a target node in the right clique, so the answer has to cross the bridge (distance $1+b$ hops) for it to work. Loss and accuracy are read at the target node, preventing any pooling shortcut. Sweep goes over depth K and bridge length b independently. 400 graphs per setting are used, in a classic 60/20/20 train/val/test split.

Rewiring. Here, an SDRF-inspired rewiring is used: each step targets the most negatively curved edge (Ollivier-Ricci, $\alpha=0$) and adds a shortcut between its low-degree neighbours. This differs from [5] mainly in the edge-selection rule (here: deterministic, degree-based, rather than softmax-curvature improvement approach); Hence it is “SDRF-inspired”. Fig. 1 shows it removes the negative-curvature bridge edges and adds additional shortcuts.

Diagnostics & protocol. Oversmoothing is measured by MAD (mean pairwise cosine distance of embeddings) and Dirichlet energy [1, 3]; over-squashing by the Jacobian sensitivity $\|\partial \mathbf{h}_v / \partial \mathbf{x}_u\|$ vs. $u-v$ distance [5, 6]. Adam is used (SBM full-batch, lr 0.01; barbell mini-batched, lr 0.005, batch 32) with early stopping on a validation split. All results are mean $\pm$ std over 5 seeds; the SBM is fixed (seeds vary initialisation), the barbell is re-drawn per seed. Main assumption is that the synthetic graphs isolate each pathology and that MAD/Jacobian are valid proxies.

3 Results

Rewiring removes the bottleneck. The barbell has exactly 4/23 negatively curved edges (== the bridge) and SDRF eliminates them, shifting the whole curvature distribution positive (App. A). This merely confirms that the intervention does what it should before going into answering the research question.

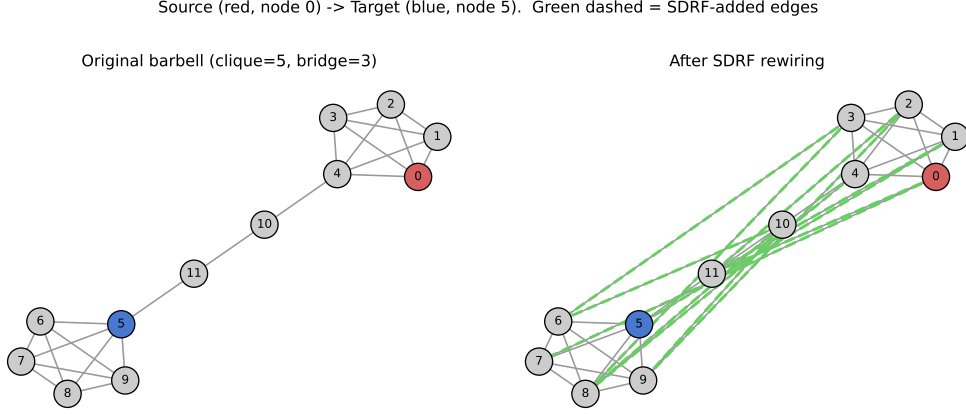


Figure 1: Barbell task before/after SDRF-inspired rewiring. Source (red) and target (blue) are separated by a thin bridge (negative curvature); rewiring (the green dashed edges) connects the clusters directly, collapsing the diameter from 5 to 2.

Experiment A - oversmoothing (Fig. 2). Plain GCN collapses with depth as expected and already tested in the practicals: accuracy $1.0 \rightarrow 0.19 \pm .03$ at $K=16$ and MAD $0.343 \pm .037 \rightarrow 0.000 \pm .000$. Skip connections substantially mitigate this (Acc $0.97 \pm .03$ at $K=8$) but do not fully solve it: at $K=16$ they are $0.86 \pm .26$ - high variance, i.e. some initialisations still fail (as could be seen in the practicals).

Most importantly, **rewiring worsens oversmoothing**: GCN+REWIRE ($0.44 \pm .15$ at $K=8$) is below GCN ($0.64 \pm .23$), with lower MAD, because the added edges increase connectivity and hence accelerate the neighbour-averaging that drives the collapse.

Experiment B - over-squashing (Fig. 3, Table 2 in App. B)). With depth (bridge $b=3$, 4 hops needed), GCN never reliably solves the task at any K (e.g. $0.57 \pm .22$ at $K=8$), whereas SKIPGCN is "perfect" for $K \geq 8$ ($1.000 \pm .000$). Going from there, sweeping bridge length from 1 to 7 at $K=8$ gives the result that under the given circumstances rewiring solves over-squashing at every bridge length ($1.000 \pm .000$ throughout), while skip connections degrades slowly but surely - perfect at $b=3$, bimodal at $b=4$ ($0.73 \pm .23$), and essentially broken from $b \geq 5$ ($0.46 \pm .04$). At $b=4-5$ skip alone starts becoming insufficient but rewiring still succeeds.

Experiment C - propagation mechanism (Fig. 4). Jacobian sensitivity decays with distance for all models, but SKIPGCN retains 100-600 $\times$ more long-range sensitivity than GCN at every distance (gap fairly robust to seed variance), explaining why it transports the signal further¹.

Summary. Table 1 consolidates the $K=8$ slice. One can say that the hypothesis is supported: the two interventions are complementary but partially asymmetric, and GCN+REWIRE is the cell that proves the trade-off is real in these circumstances - best on the over-squashing axis, worst on the oversmoothing axis.

¹Rewired graphs have datapoints only up to distance 2: SDRF shrank the diameter so no longer-range pairs exist - itself direct evidence the bottleneck is gone.

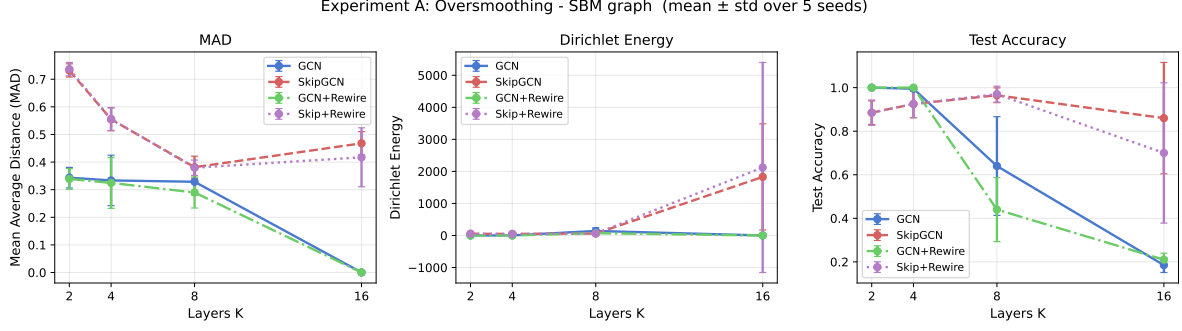


Figure 2: Experiment A (SBM, mean $\pm$ std over 5 seeds). GCN collapses with depth (MAD $\rightarrow$ 0, Acc $\rightarrow$ chance); skip connections mitigate but show large variance at $K=16$; rewiring lowers MAD and accuracy relative to its non-rewired counterpart.

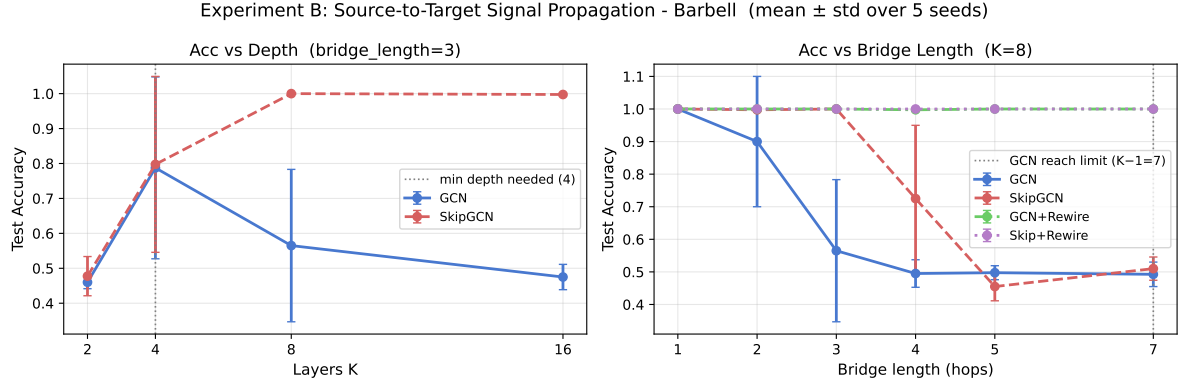


Figure 3: Experiment B (barbell, mean $\pm$ std over 5 seeds). Left: accuracy vs. depth ($b=3$). Right: accuracy vs. bridge length ($K=8$). Rewired conditions are at 1.0 throughout; SKIPGCN fails for $b \geq 5$.

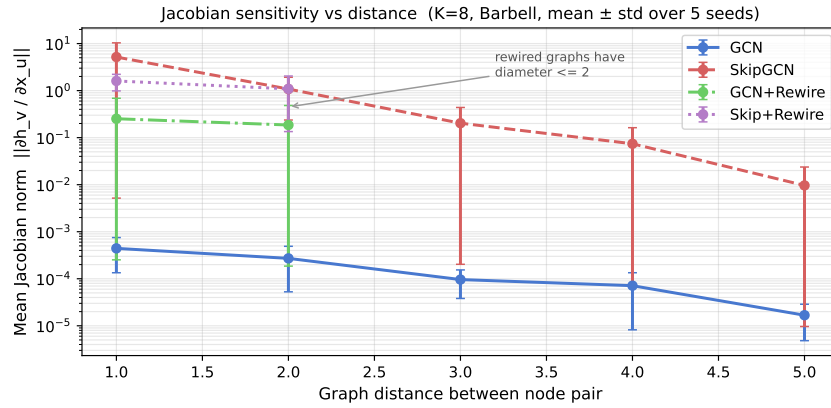


Figure 4: Experiment C: Jacobian sensitivity vs. distance ($K=8$, mean $\pm$ std over 5 seeds). SKIPGCN retains far more long-range sensitivity; SRDF rewired curves stop at distance 2 because the diameter shrinks.

Table 1: Summary at $K=8$ (mean $\pm$ std over 5 seeds).

Model	MAD (SBM)	Acc (SBM)	Acc (Barbell $b=3$)
GCN	.329 $\pm$.047	.640 $\pm$.227	.565 $\pm$.218
SKIPGCN	.381 $\pm$.040	.965 $\pm$.034	1.000$\pm$.000
GCN+REWIRE	.290 $\pm$.056	.440 $\pm$.147	1.000$\pm$.000
SKIP+REWIRE	.379 $\pm$.028	.970 $\pm$.037	1.000$\pm$.000

4 Discussion

Critical perspective. There are a few things that can and should be improved upon in the future as this ”miniproject” was done in a limited scope and time:

- (a) Rewiring ”wins” partly by changing the task: SDRF collapses the source-target distance to 1-2 hops, so at $K=8$ the problem becomes trivial - the perfect $1.000 \pm .000$ reflects an easier graph, not a model that squashes less.
- (b) The barbell metric is bimodal (the runs either solve it, ≈ 1.0 , or sit at chance, ≈ 0.5); a mean of 0.73 means roughly ”3/5 seeds solved it,” not a typical 73%.
- (c) Skip connections are a *mitigation, not a cure*: at $K=16$ ($.86 \pm .26$) and $b=4$ ($.73 \pm .23$) they are unreliable. A note that was already taken in the practicals though.
- (d) The Jacobian’s absolute scale is unstable across seeds (e.g. SkipGCN 5.2 ± 5.2 at distance 1); At least the decay and the gaps can be seen as useful.
- (e) Last but not least: The datasets are small and synthetic (by design); they are supposed to isolate the mechanisms but obviously do not provide results on real and more complex graphs.

5 Conclusions and Outlook

In this controlled 2×2 study we see that skip connections and curvature-based rewiring are complementary but asymmetric: rewiring seems to fix the over-squashing but worsens over-smoothing; skip connections mitigate over-smoothing and moderate over-squashing but fail on longer bottlenecks, and only their combination is robust on both. Practically, ”stacking fixes” seem to be justified here, but each fix has a cost on the opposite side.

Outlook. To study this in more detail following points are to be considered in the future:

- (i) Test on real(life) long-range benchmarks to see whether the trade-off persists beyond synthetic graphs;
- (ii) Fully implemented SDRF (softmax-sampled by curvature gain) and alternative rewiring strategies to separate the current ”shortening the path” from ”actually changing the curvature”;
- (iii) a graph dataset where rewiring cannot trivially shorten the source-target distance; essentially to test rewiring’s benefit when it does not change the task difficulty.

References

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A Curvature distribution

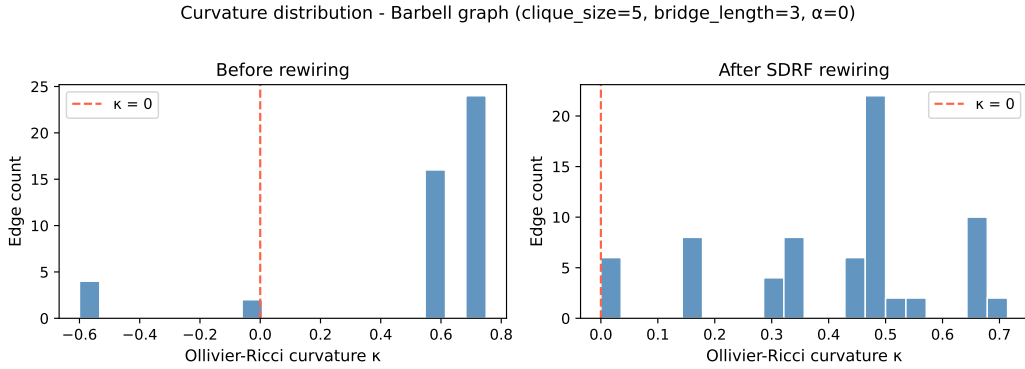


Figure 5: Ollivier-Ricci curvature of the barbell before/after rewiring. The negative (bottleneck) edges are removed and the distribution shifts positive.

B Oversquashing over bridge-length

Table 2: Over-squashing: target-node accuracy vs. bridge length ($K=8$, mean $\pm$ std over 5 seeds). Rewiring is invariant to bridge length; skip connections break at $b \geq 5$.

b (hops)	1	2	3	4	5	7
GCN	1.00	.90 $\pm$.20	.57 $\pm$.22	.50 $\pm$.04	.50 $\pm$.02	.49 $\pm$.04
SKIPGCN	1.00	1.00	1.00	.73 $\pm$.23	.46 $\pm$.04	.51 $\pm$.04
GCN+REWIRE	1.00	1.00	1.00	1.00	1.00	1.00
SKIP+REWIRE	1.00	1.00	1.00	1.00	1.00	1.00